

Figure 4: Adaptive Therapy via the ABC and VEGF Models

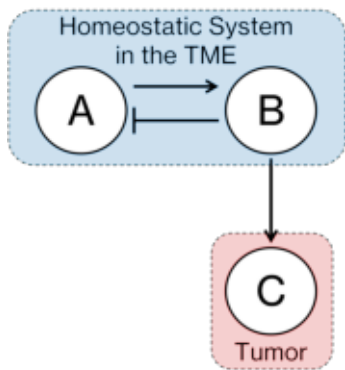
A

ABC Model

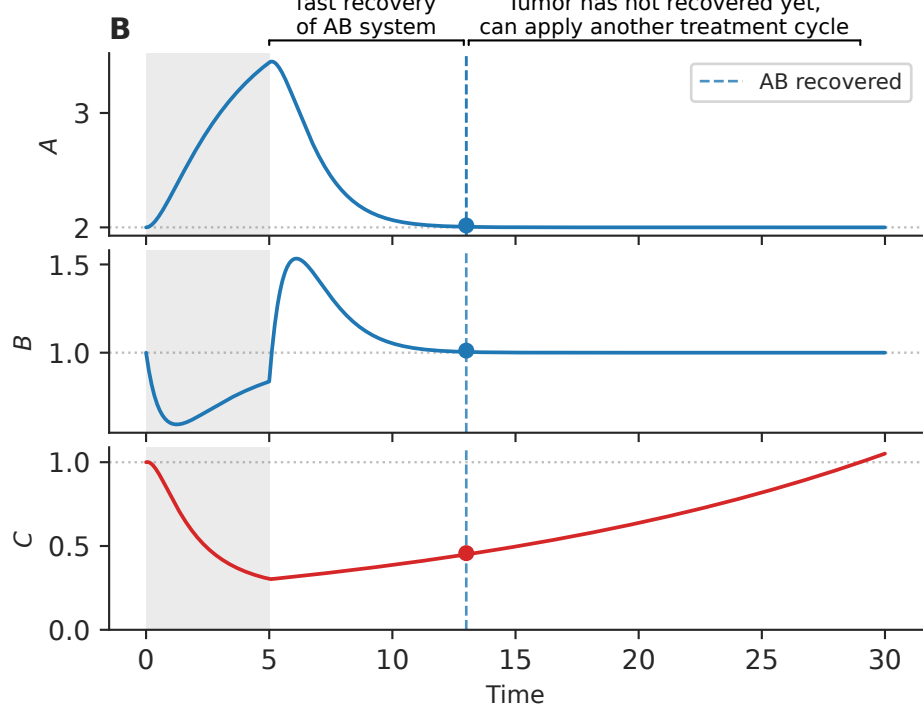
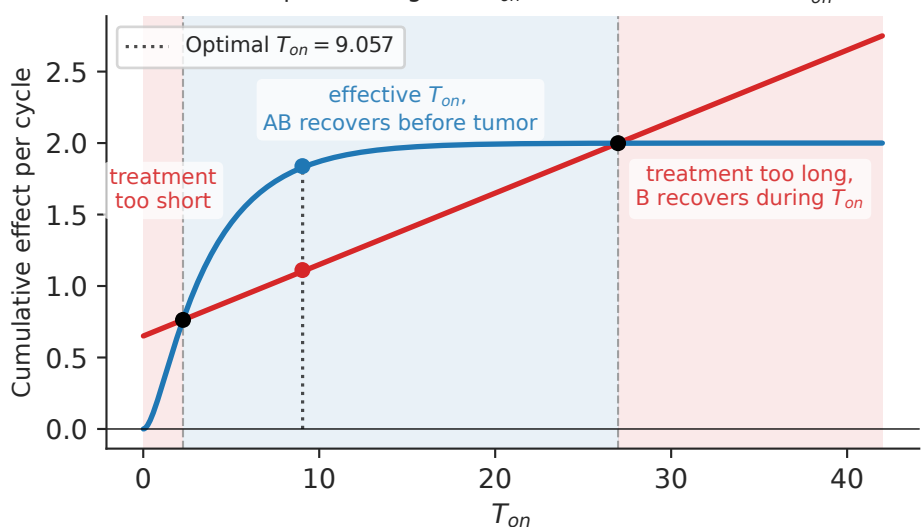
$$\frac{dA}{dt} = a_1(\mu - B)$$

$$\frac{dB}{dt} = b_1A - b_2B$$

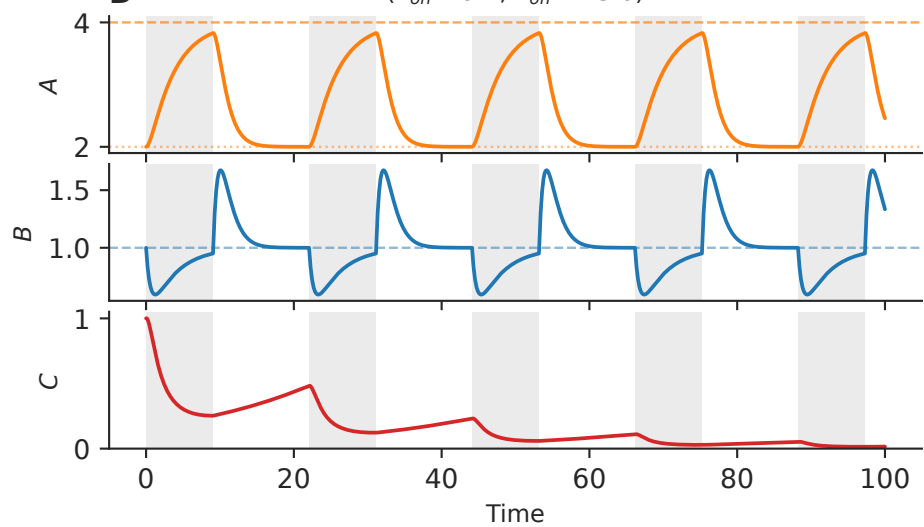
$$\frac{dC}{dt} = C(c_1 - c_2(\mu - B)^+)$$



Timescale separation: the homeostatic system recovers before the tumor

Comparing tumor growth and decay in the ABC model, **C** to find the optimal range for T_{on} $T_{off} = 13.0$ 

Adaptive therapy for the optimal treatment time found

 $(T_{on} = 9.1, T_{off} = 13.0)$ **E**

Angiogenesis Model

Based on ABC model principles, augmented with redundancy and feedback

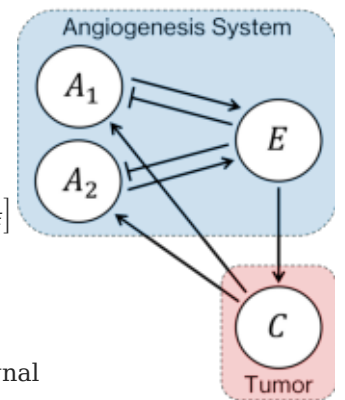
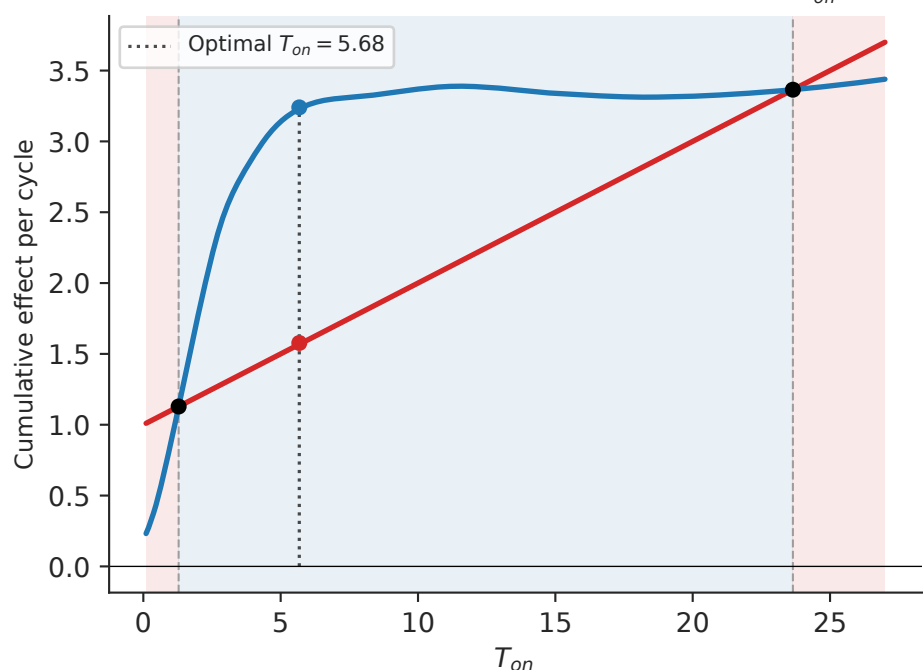
$$\frac{dA_1}{dt} = k_1(\mu C - E)$$

$$\frac{dA_2}{dt} = k_2(\mu C - E)$$

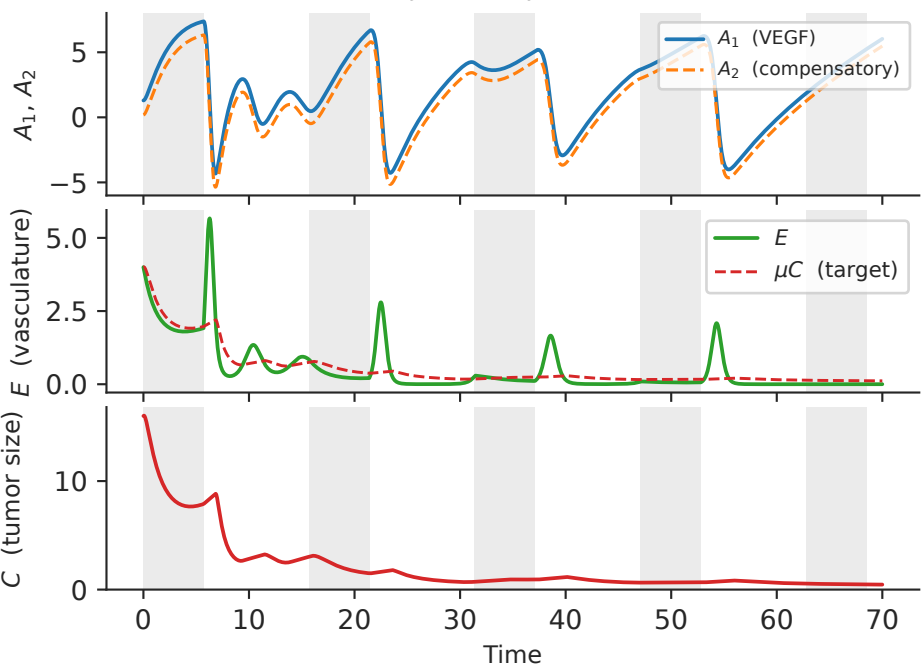
$$\frac{dE}{dt} = E \left[(a_1A_1 + a_2A_2) \left(1 - \frac{E}{4\mu C_0} \right) - r_E \right]$$

$$\frac{dC}{dt} = C(c_1 - c_2(\mu C - E)^+)$$

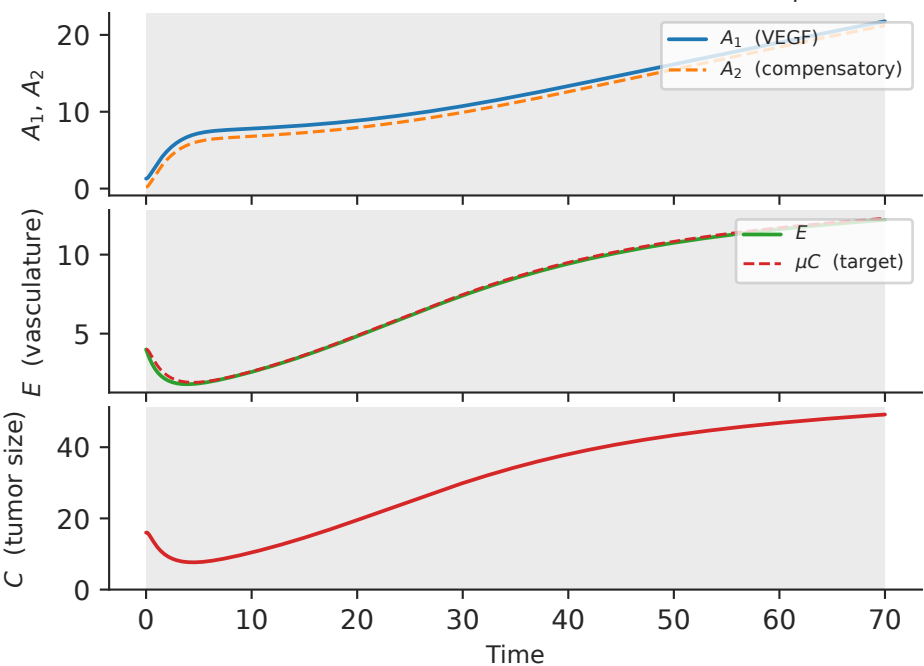
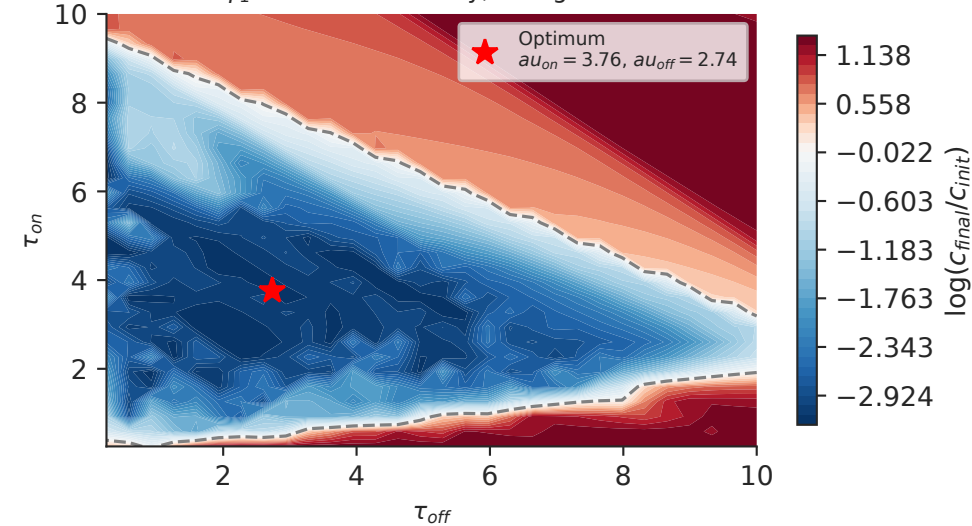
A_1 : VEGF, A_2 : compensatory signal
 E : vasculature, C : tumor size

Comparing tumor growth and decay in the VEGF model, **F** to find the optimal range for T_{on} $T_{off} = 10.0$ 

Adaptive therapy for the optimal treatment time found

 $(T_{on} = 5.7, T_{off} = 10.0)$ 

Continuous treatment (VEGF model) - tumor escapes

Tumor log-fold change $\log(c_{final}/c_{init})$ over (τ_{on}, τ_{off}) $\gamma_1 = 0.3$ - blue: decay, red: growthTumor log-fold change $\log(c_{final}/c_{init})$ over (τ_{on}, τ_{off}) $\gamma_1 = 1.3$ - blue: decay, red: growth